

Disclaimer

Thank you for your interest in using Google Cloud training material. We are excited about the content we are able to provide you (collectively, “Teaching Resources”) and hope you are too.

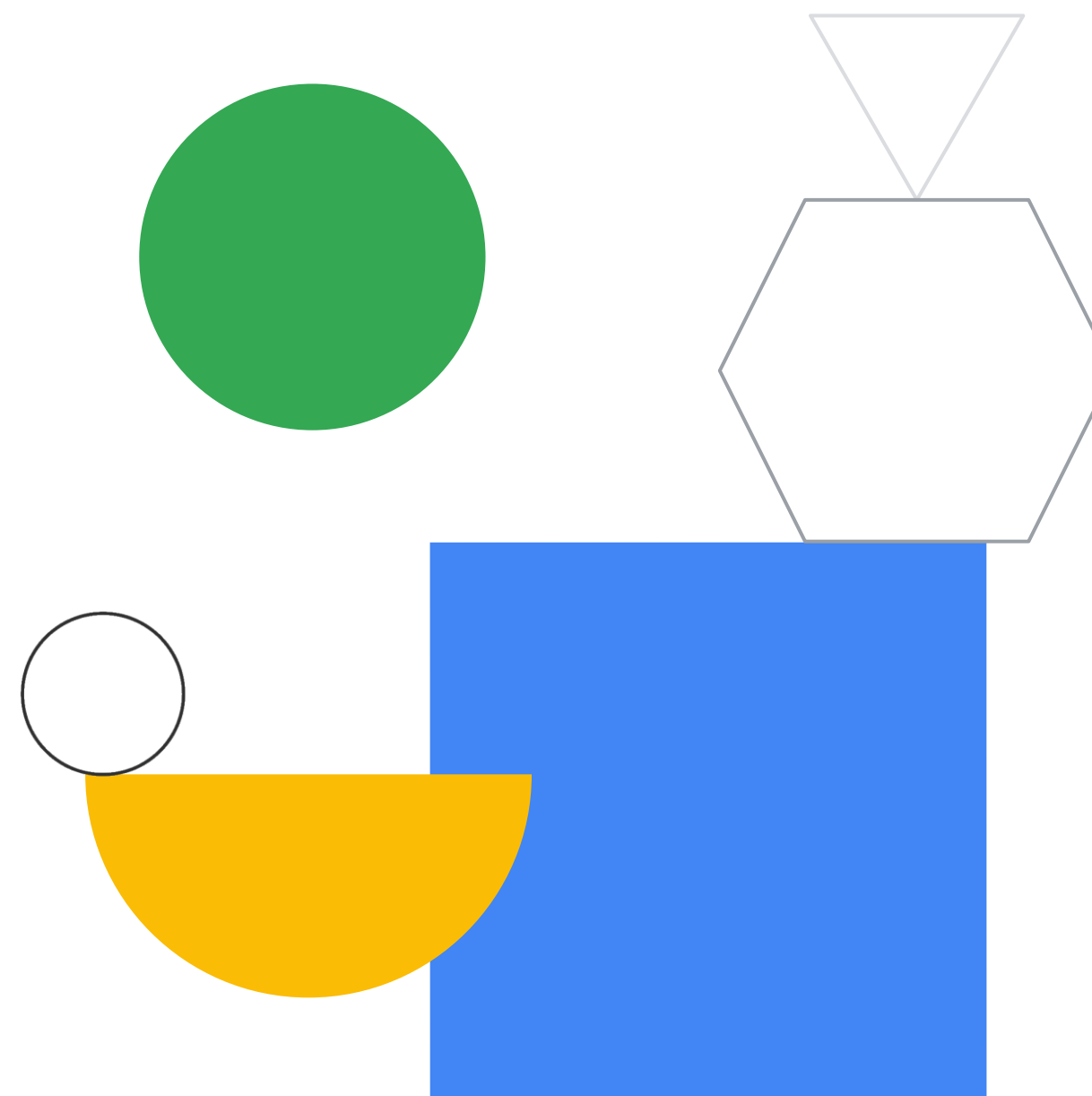
By using Teaching Resources, you agree to be bound by all of the following terms, as well as the [Google terms of service](#) and the [Google privacy policy](#). Unless otherwise specified, terms used below will have the meanings described in the [Google terms of service](#).

1. **Educational Use Only.** Teaching Resources is intended only for use teaching courses at regionally accredited or higher learning institutions. The content may be adapted, customize, remixed, and shared for educational use. Content may not be distributed, or otherwise exploited for any commercial purpose, commercial advantage, or private monetary compensation.
1. **Attribution Requirements.** If you distribute, publicly perform, display, transmit, publish, or otherwise make available any Teaching Resources or any derivative works thereof, you must attribute the material you use back to the Teaching Resources, but not in any way that suggests Google, any of its affiliates, or any of its third party content providers endorse you or your use of such materials. If you adapt, remix, or customize the Teaching Resources please include the following text on each edited slide: *“The original content was provided by Google LLC and modified for the purpose of the course, without input or endorsement from Google LLC.”*
1. Descriptions of Google products, services, infrastructure and processes contained in the Teaching Resources are descriptions for teaching and learning purposes only and do not constitute a guarantee, promise, or representation of accuracy by Google. Pricing, availability or features of Google Cloud products and services described in the Teaching Resources may change.



You can't secure the cloud, right?

Module 6 - Security
Google Cloud Computing Foundations



Course map

01 —

So, what's the cloud anyway?

02 —

Start with a solid platform

03 —

Use Google Cloud to build your apps

04 —

Where do I store this stuff?

05 —

There's an API for that!

06 —

You can't secure the cloud, right?

07 —

It helps to network

08 —

Keeping an eye on things

09 —

You have the data, but what are you doing with it?

10 —

Let machines do the work

Google Cloud skill badges

Objectives

- 01 Outline how security is administered in Google Cloud.
- 02 Explore the different encryption options with Google Cloud.
- 03 Identify best practices when you configure authentication and authorization with Identity and Access Management (IAM).





Agenda



- 01 Security in the cloud
- 02 The shared security model
- 03 Encryption options
- 04 Authentication and authorization with IAM
- 05 Lab: User Authentication: Identity-Aware Proxy
- 06 IAM authorization best practices
- 07 Lab: IAM: Qwik Start
- 08 Quiz
- 09 Summary

01 Security in the cloud

02 The shared security model

03 Encryption options

04 Authentication and authorization with IAM

05 Lab: User Authentication: Identity-Aware Proxy

06 IAM authorization best practices

07 Lab: IAM: Qwik Start

08 Quiz

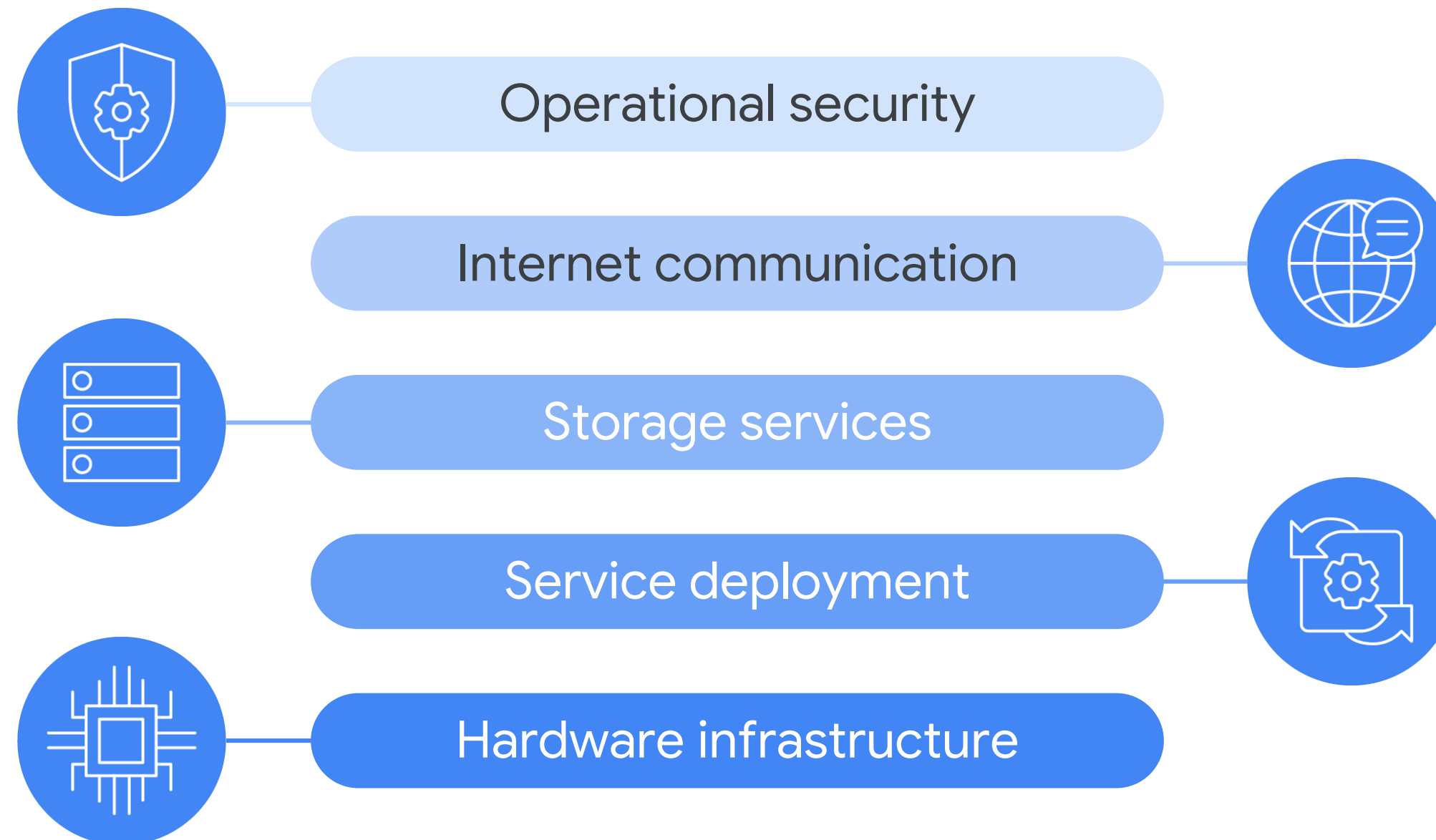
09 Summary

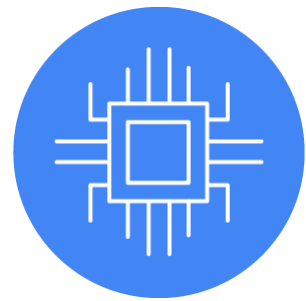
Agenda

Google Cloud is
designed with
security in mind



Infrastructure security design





Hardware infrastructure layer

Hardware design and provenance

Both the server boards and the networking equipment in Google data centers are custom designed by Google. Google also designs custom chips, including a hardware security chip.

Secure boot stack

Google's server machines use a variety of technologies to ensure that they are booting the correct software stack.

Premises security

Google designs and builds its own data centers, which incorporate multiple layers of physical security protections.



Service deployment layer

Encryption of inter-service communication

Google's services communicate with each other using remote procedure calls ("RPCs").

Google's infrastructure provides cryptographic privacy and integrity by automatically encrypting all infrastructure RPC traffic between data centers.

User identity

Google's central identity service intelligently challenges users for additional information based on certain risk factors.

Users can use secondary factors when signing in, including devices based on the Universal 2nd Factor (U2F) open standard.



Storage services layer

Encryption at rest

Most Google applications access file storage indirectly via storage services and encryption.

Using centrally managed keys, encryption is applied at the layer of these storage services.

Google also enables hardware encryption support in hard drives and SSDs.



Internet communication layer

Google Front End (“GFE”)

Google Front End ensures that all registered services use TLS connections incorporating the correct certificates and following best practices.

Denial of Service (“DoS”) protection

Google has multi-tier, multi-layer DoS protections that reduce the risk of any DoS impact on
a service running behind a GFE



Operational security layer

Intrusion detection

Rules and machine intelligence give operational security engineers warnings of possible incidents.

Reducing insider risk

Google limits and monitors the activities of employees who have access to the infrastructure.

Employee Universal Second Factor (U2F) use

All Google employee accounts require use of U2F-compatible Security Keys.

Software development practices

Google employs central source control and requires two-party review of any new code.



Agenda



- 01 Security in the cloud
- 02 The shared security model**
- 03 Encryption options
- 04 Authentication and authorization with IAM
- 05 Lab: User Authentication: Identity-Aware Proxy
- 06 IAM authorization best practices
- 07 Lab: IAM: Qwik Start
- 08 Quiz
- 09 Summary

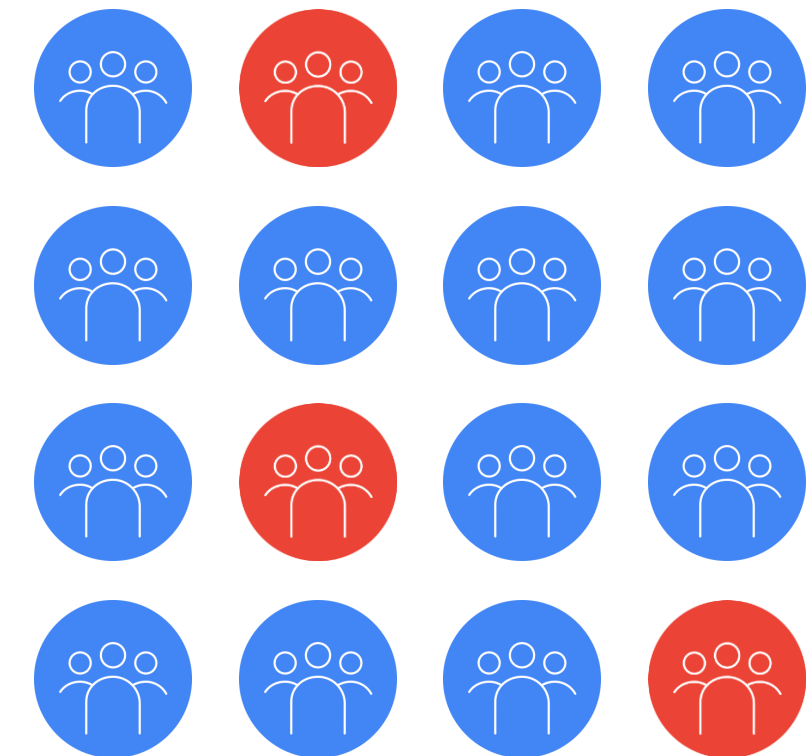
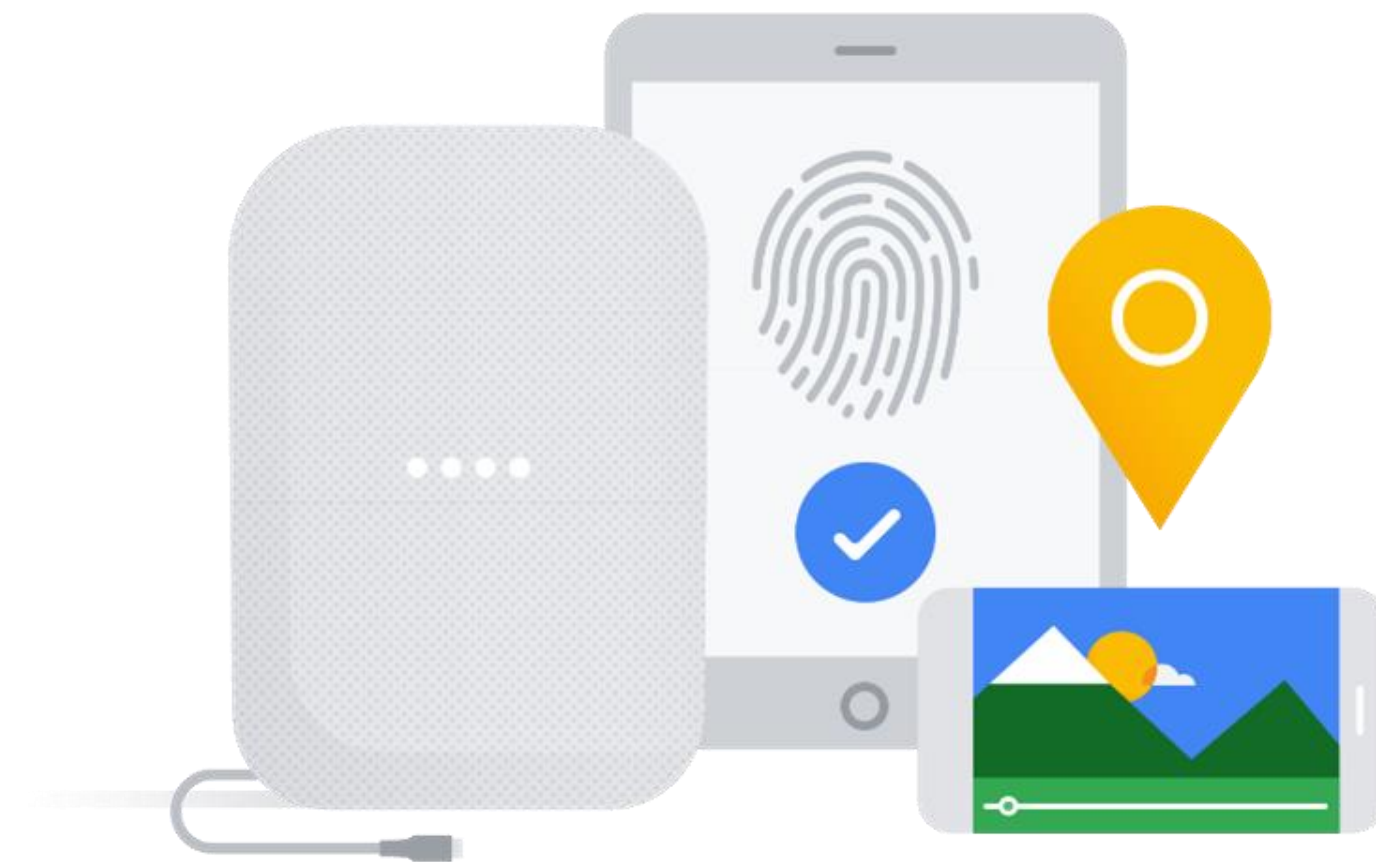
The shared security model

Security responsibilities are shared between the customer and Google Cloud.

- Customer-managed
- Google-managed

Responsibility	On-premises	IaaS	PaaS	Managed services
Content	Customer-managed	Customer-managed	Customer-managed	Customer-managed
Access policies	Customer-managed	Customer-managed	Customer-managed	Customer-managed
Usage	Customer-managed	Customer-managed	Customer-managed	Customer-managed
Deployment	Customer-managed	Customer-managed	Customer-managed	Google-managed
Web app security	Customer-managed	Customer-managed	Customer-managed	Google-managed
Identity	Customer-managed	Customer-managed	Google-managed	Google-managed
Operations	Customer-managed	Google-managed	Google-managed	Google-managed
Access and authentication	Customer-managed	Google-managed	Google-managed	Google-managed
Network security	Customer-managed	Google-managed	Google-managed	Google-managed
OS, data, and content	Customer-managed	Google-managed	Google-managed	Google-managed
Audit logging	Customer-managed	Google-managed	Google-managed	Google-managed
Network	Customer-managed	Google-managed	Google-managed	Google-managed
Storage and encryption	Customer-managed	Google-managed	Google-managed	Google-managed
Hardware	Customer-managed	Google-managed	Google-managed	Google-managed

Data access is usually the customer's responsibility





Agenda



- 01 Security in the cloud
- 02 The shared security model
- 03 Encryption options**
- 04 Authentication and authorization with IAM
- 05 Lab: User Authentication: Identity-Aware Proxy
- 06 IAM authorization best practices
- 07 Lab: IAM: Qwik Start
- 08 Quiz
- 09 Summary

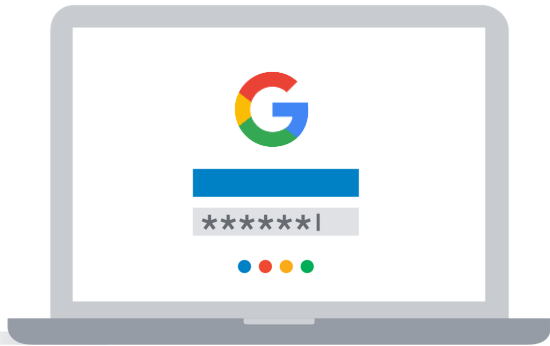
Google Cloud encryption options

Limited control
Simpler

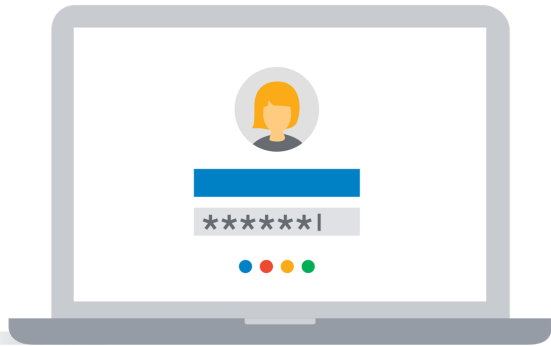
More control
More complex



Encryption by default
(only on Google Cloud)

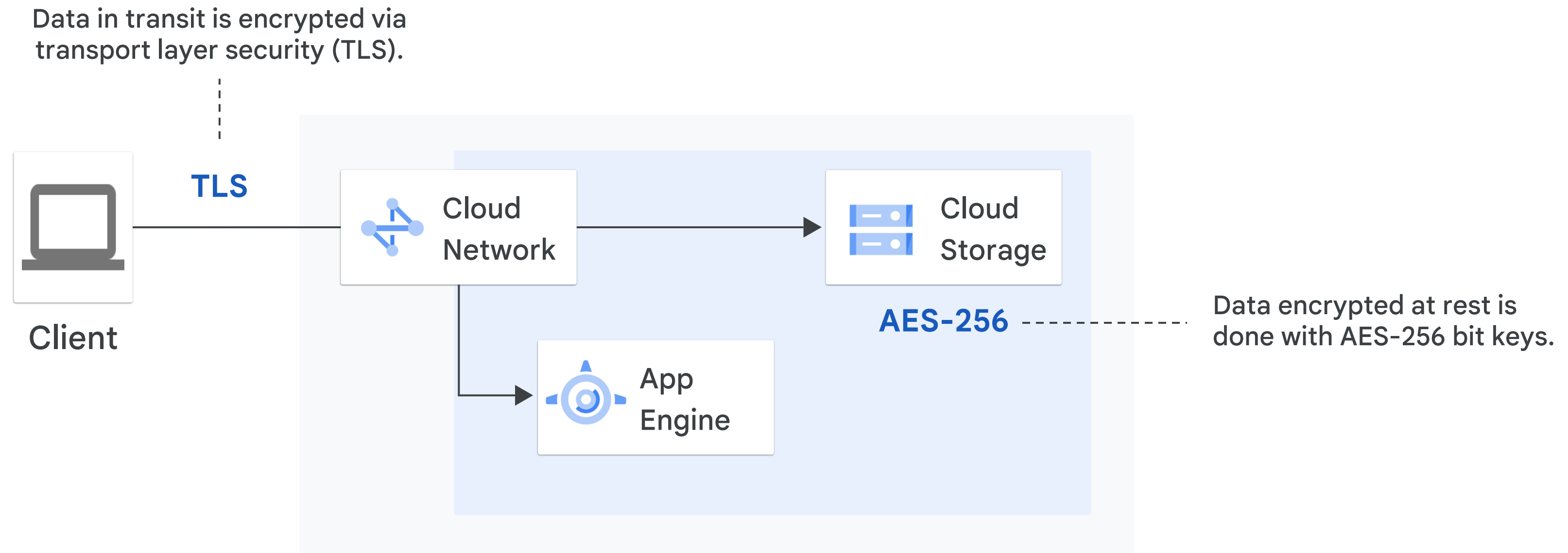


Customer-managed
encryption keys (CMEK)

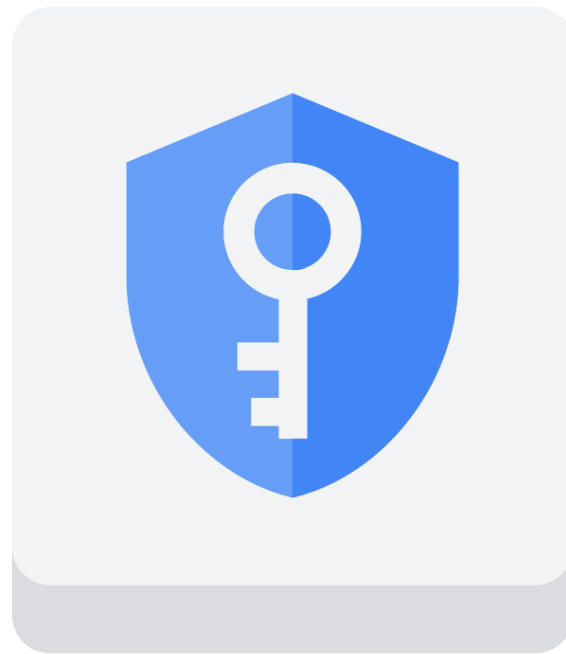


Customer-supplied
encryption keys (CSEK)

Google Cloud will encrypt data in transit and at rest by default



Customer-managed encryption keys (CMEK)



Cloud KMS

- ✓ Manage keys in a cloud-hosted solution
- ✓ Encrypt and decrypt via API
- ✓ Automated and at-will key rotation
- ✓ Symmetric and asymmetric key support

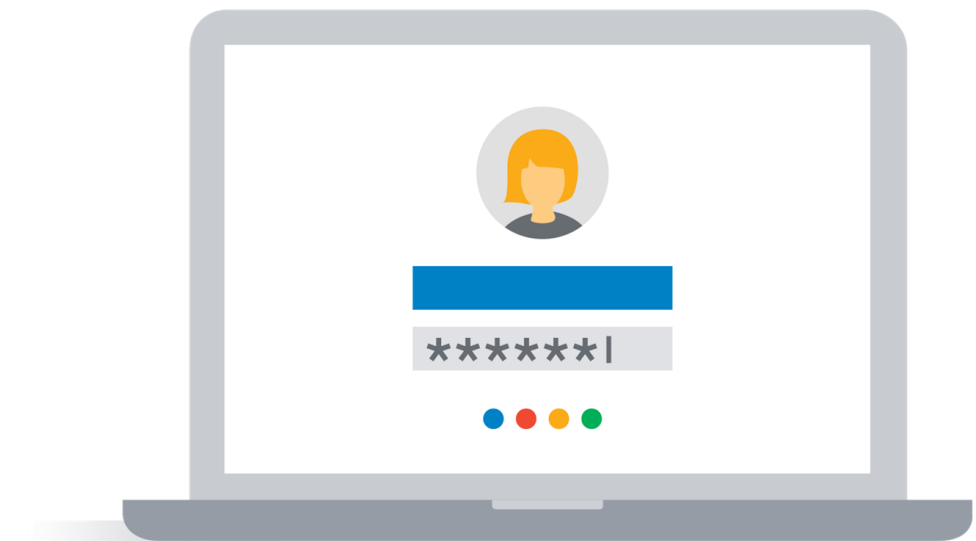
Customer-supplied encryption keys (CSEK)

Gives users more control over their keys, but with greater management complexity.

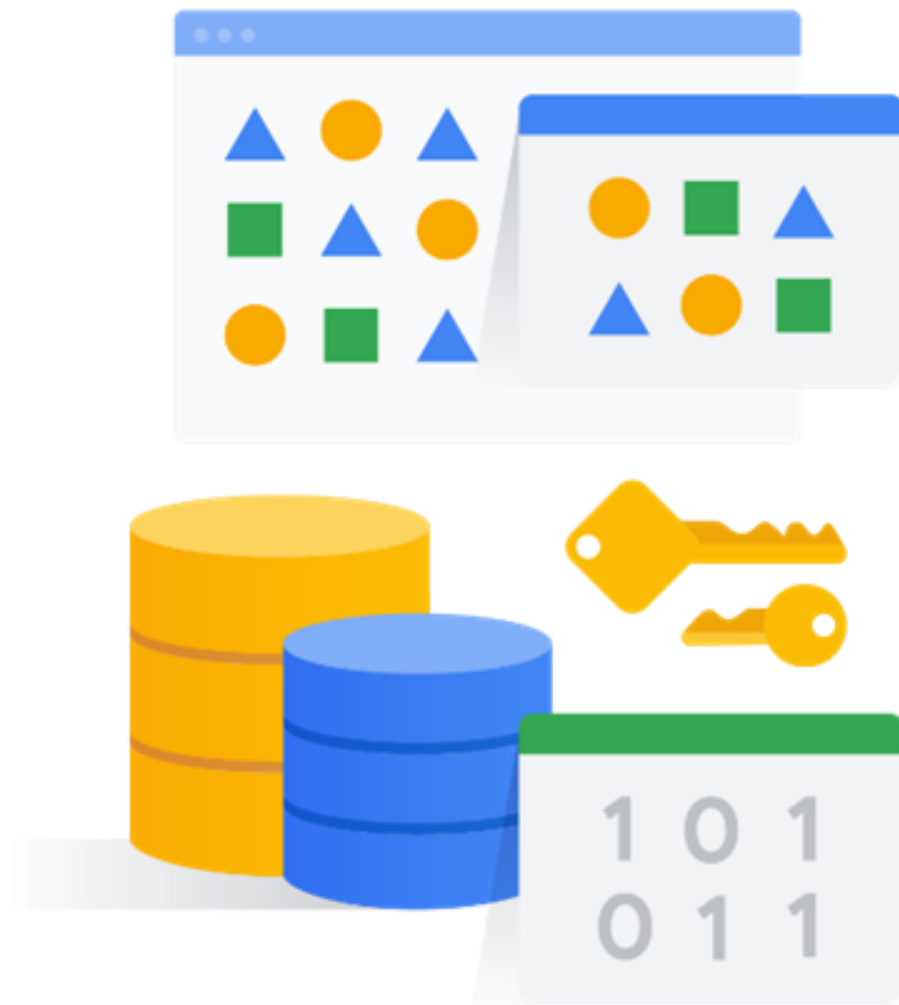
Users use their own AES-256-bit encryption keys and are responsible for generating them.

Users are responsible for storing the keys and providing them as part of Google Cloud API calls.

Google Cloud will use the provided key to encrypt the data before saving it.



Persistent disk encryption with CSEK



- ✓ Data is encrypted before it leaves the instance.
- ✓ System defined or customer supplied keys.
- ✓ The deletion process renders data irrecoverable.

Other encryption options



Disk encryption



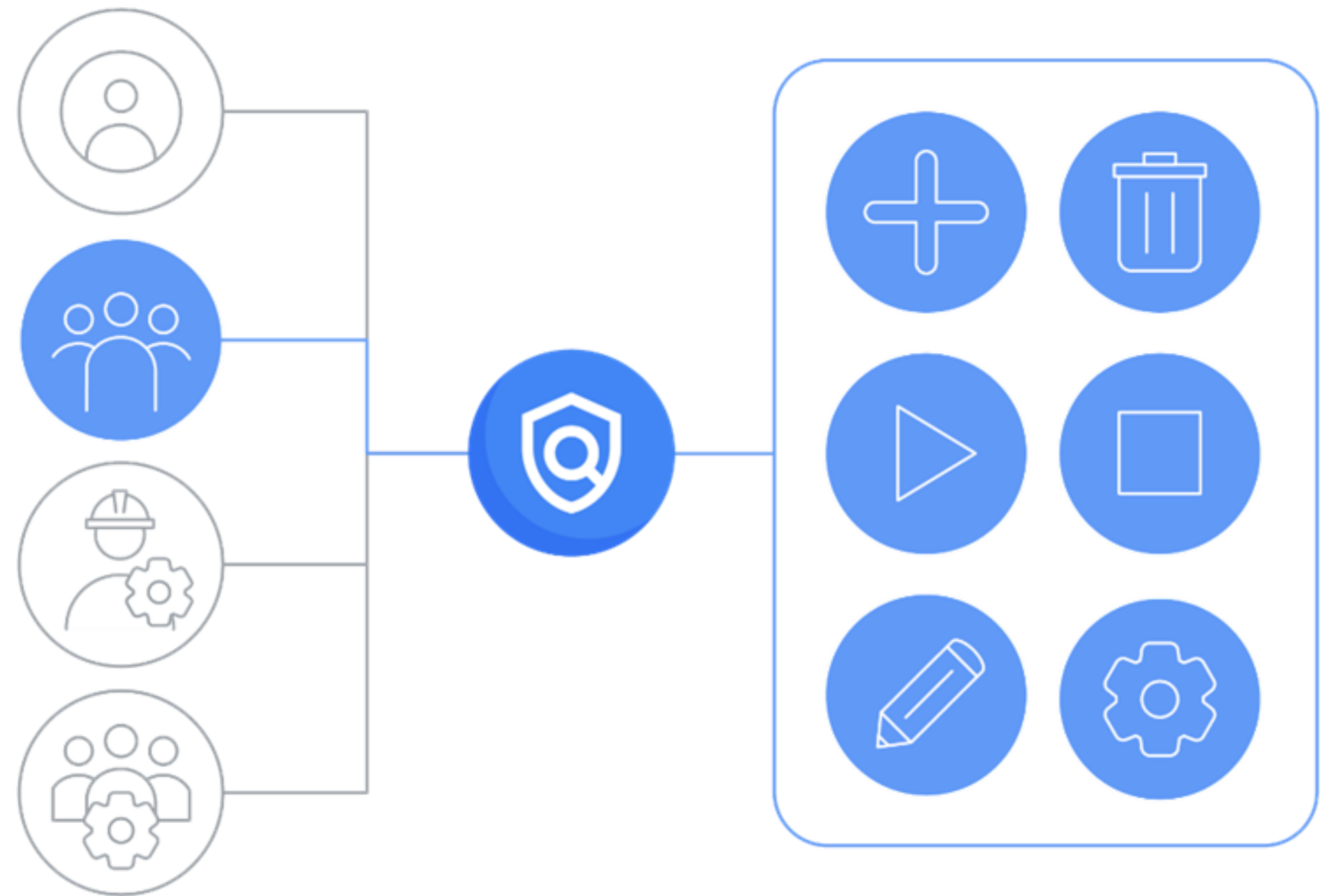
Client-side encryption

- 01 Security in the cloud
- 02 The shared security model
- 03 Encryption options
- 04 Authentication and authorization with IAM**
- 05 Lab: User Authentication: Identity-Aware Proxy
- 06 IAM authorization best practices
- 07 Lab: IAM: Qwik Start
- 08 Quiz
- 09 Summary

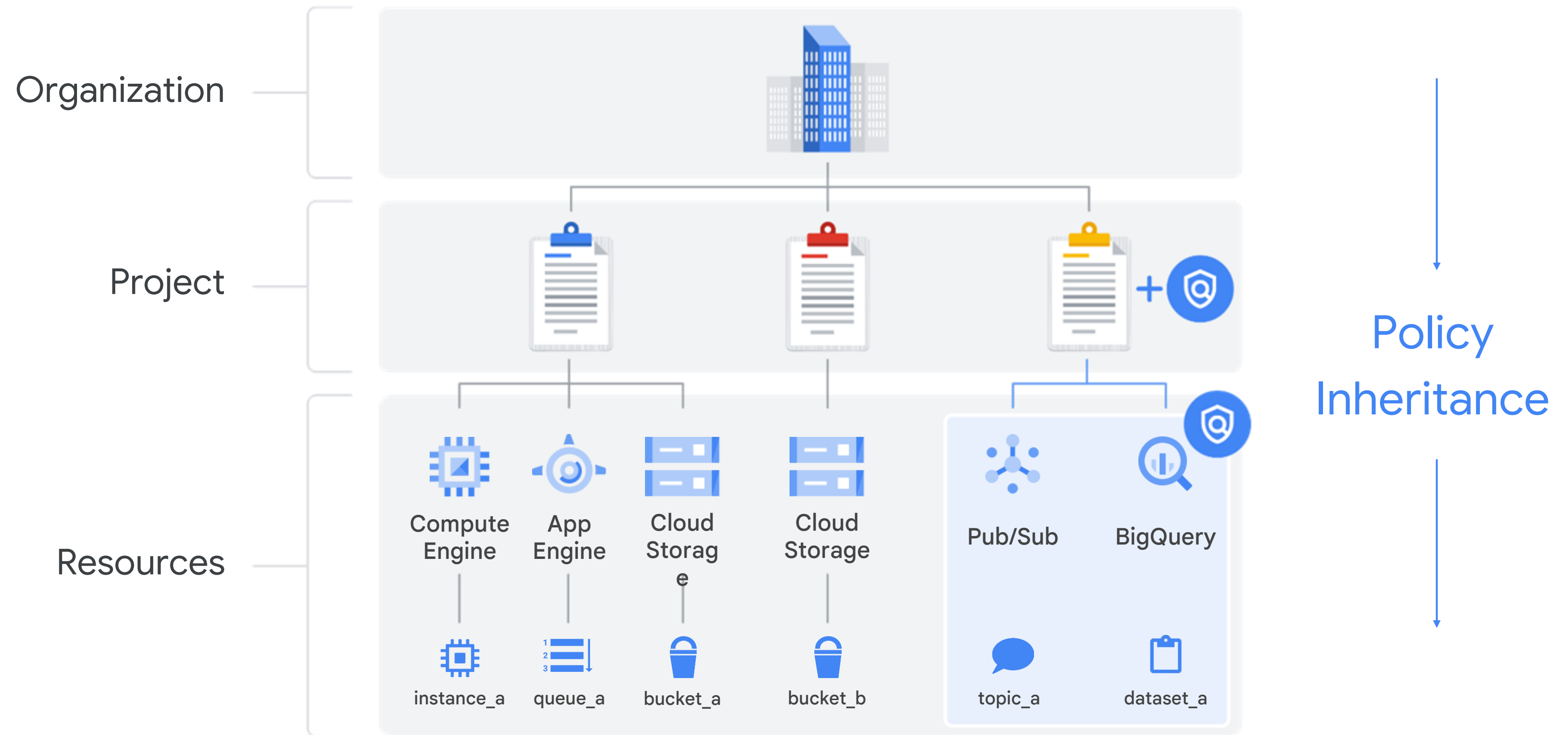
Agenda

Identity and Access Management applies policies

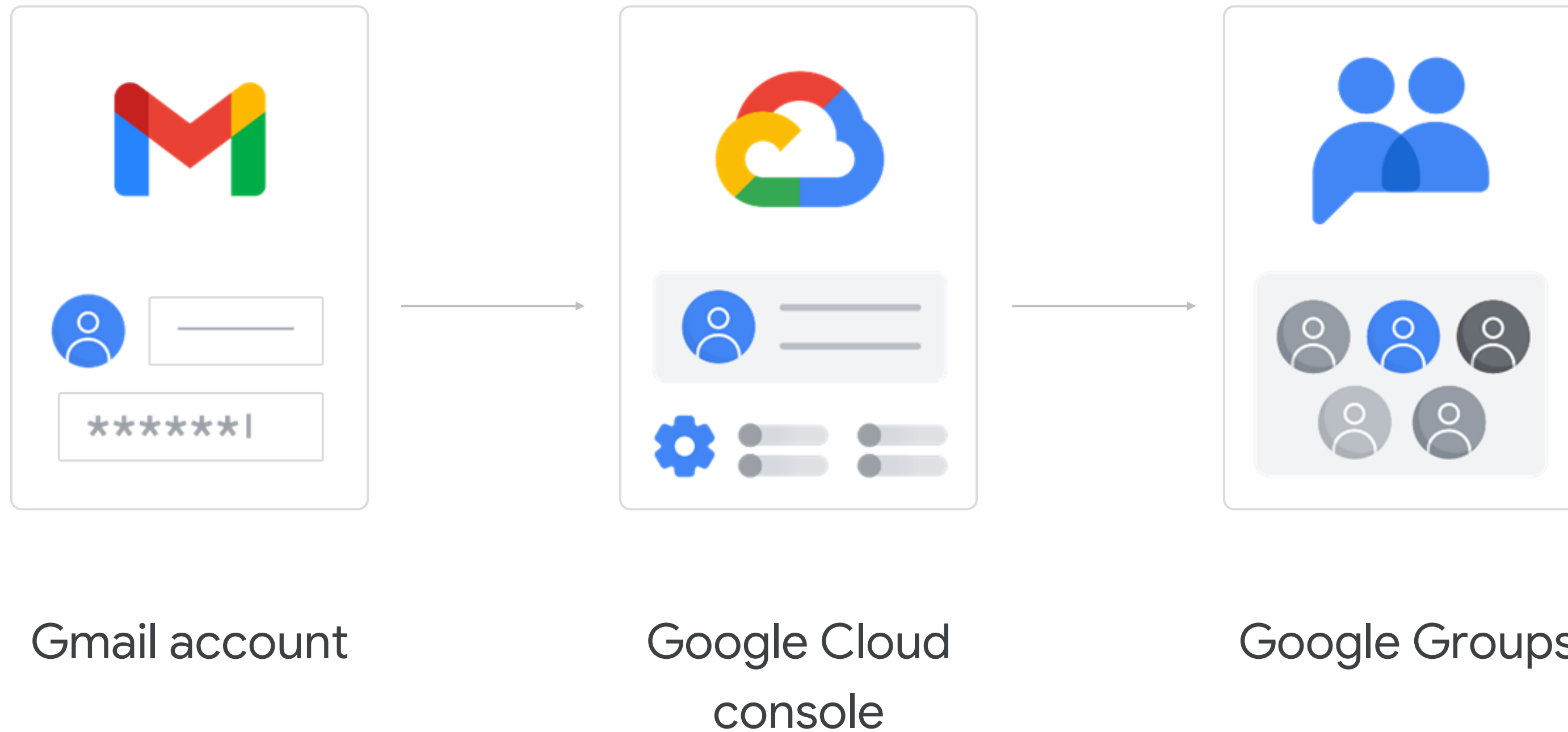
Administrators can
apply policies that
define **who** can do **what**
on **which** resources



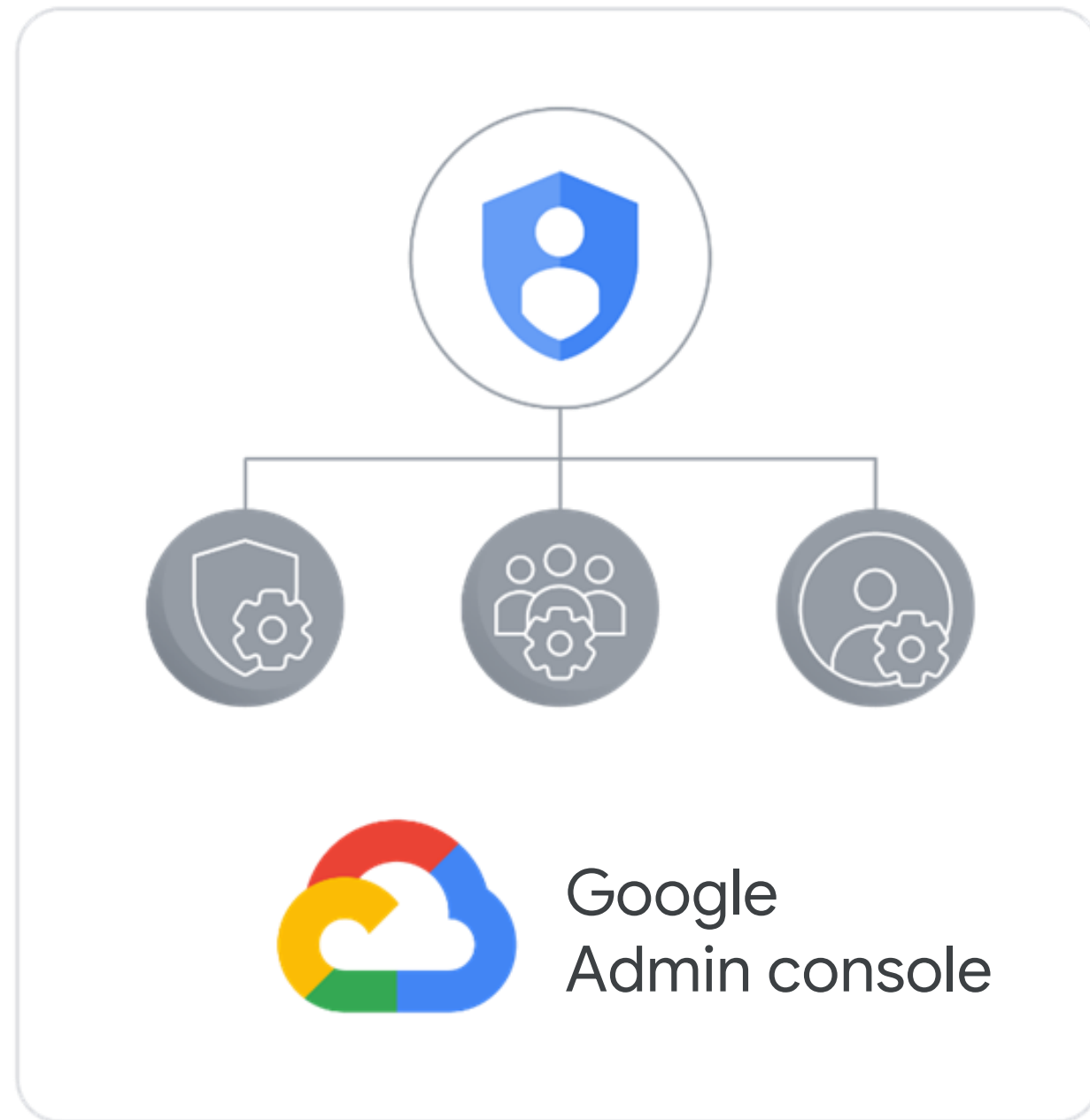
Policies are managed and applied by IAM



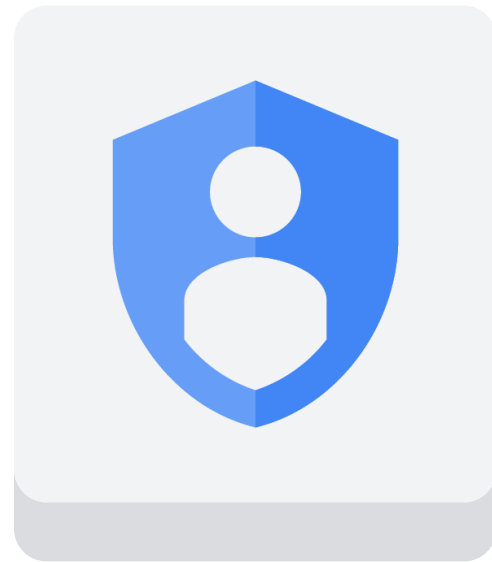
Cloud Identity manages team and organization access



Cloud Identity defines user and group policies



With [Cloud Identity](#), organizations can define policies and manage their users and groups using the [Google Admin console](#)



Cloud Identity



Log in and manage resources using the same credentials used in existing Active Directory or LDAP systems



The Google Admin console can be used to disable user accounts and remove them from groups when they leave



Available in free and premium editions



Already available to Google Workspace customers in the Google Admin console

There are three kinds of IAM roles

Basic
IAM role



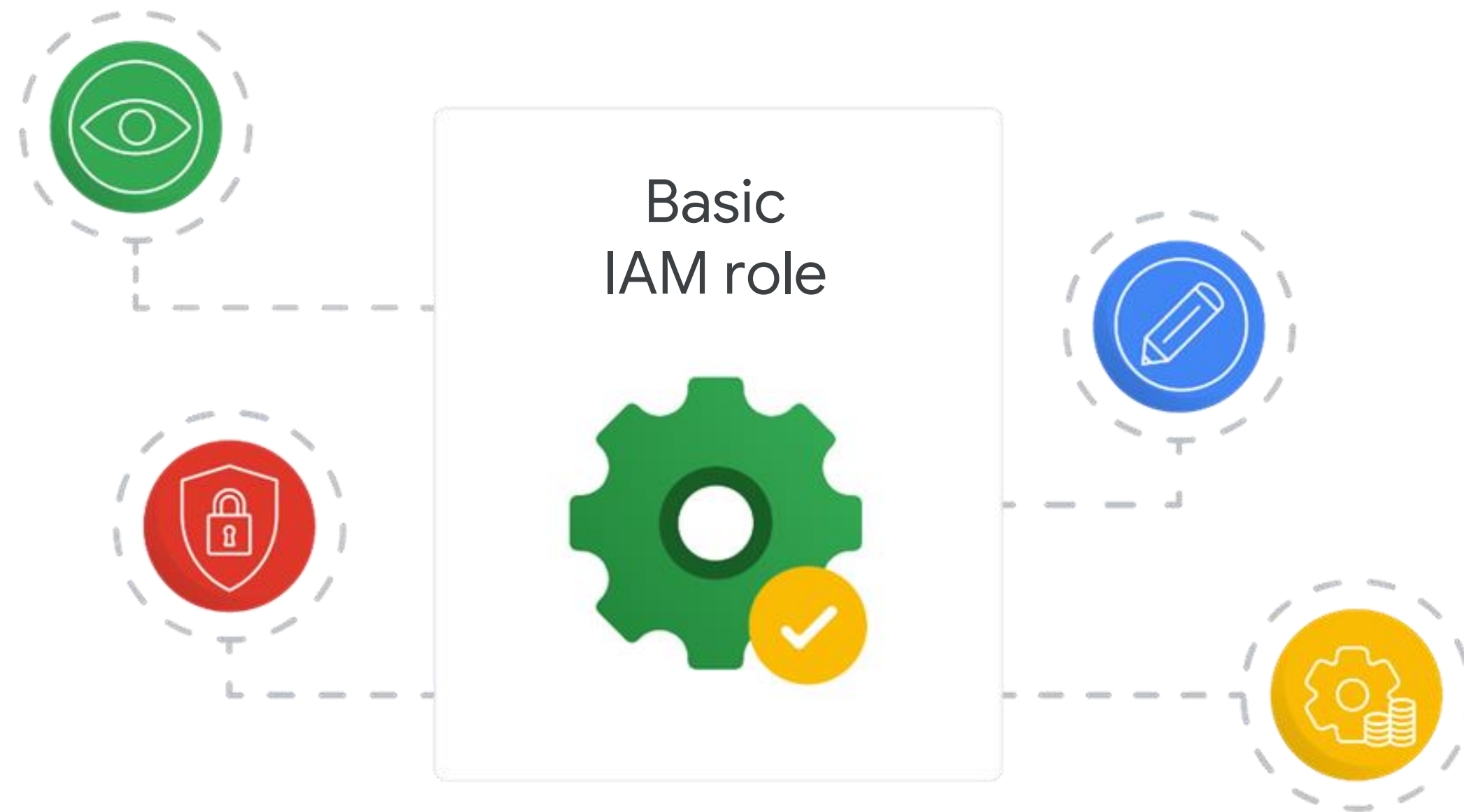
Predefined
IAM role



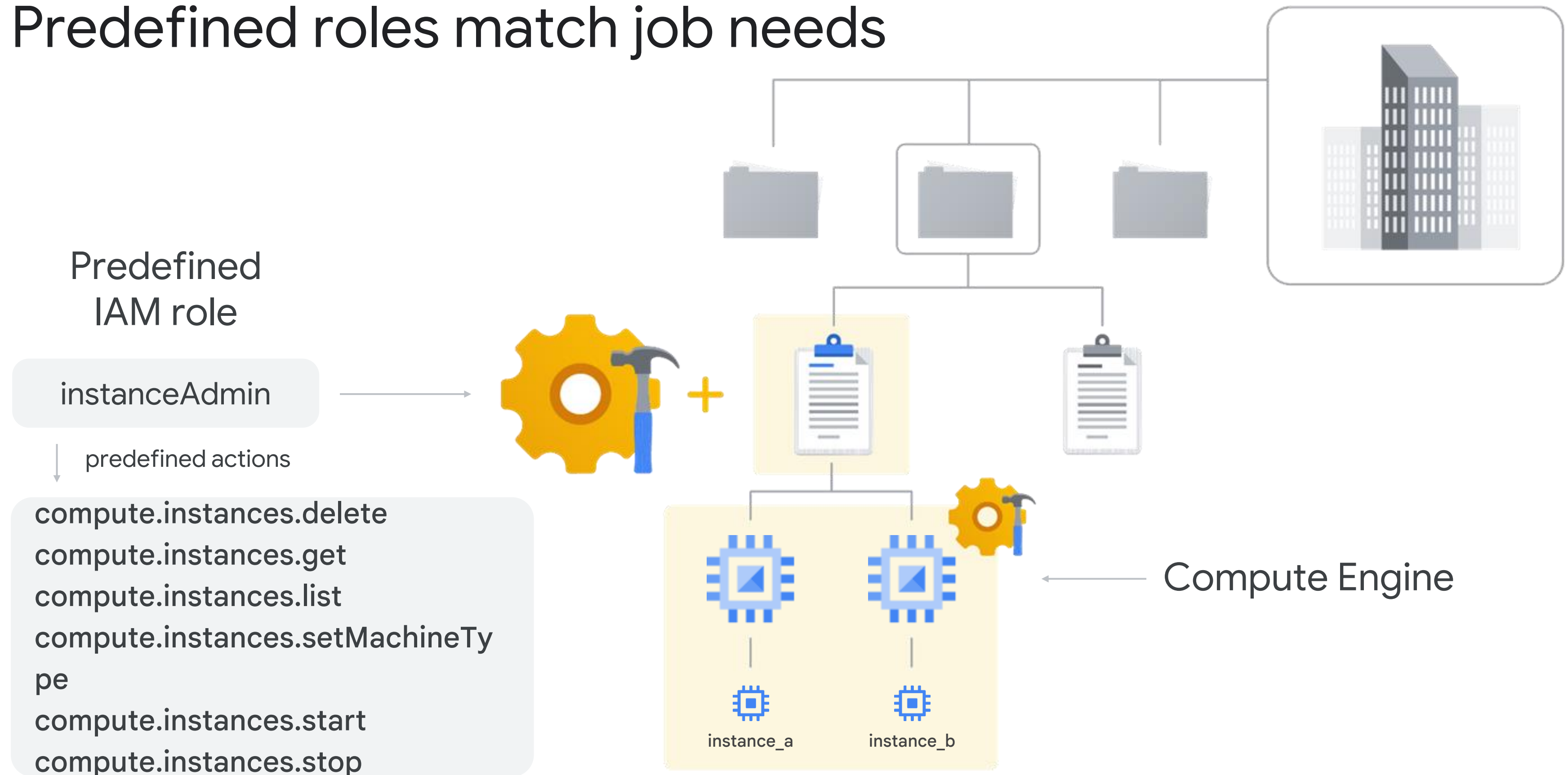
Custom
IAM role



Basic IAM roles are broad in scope



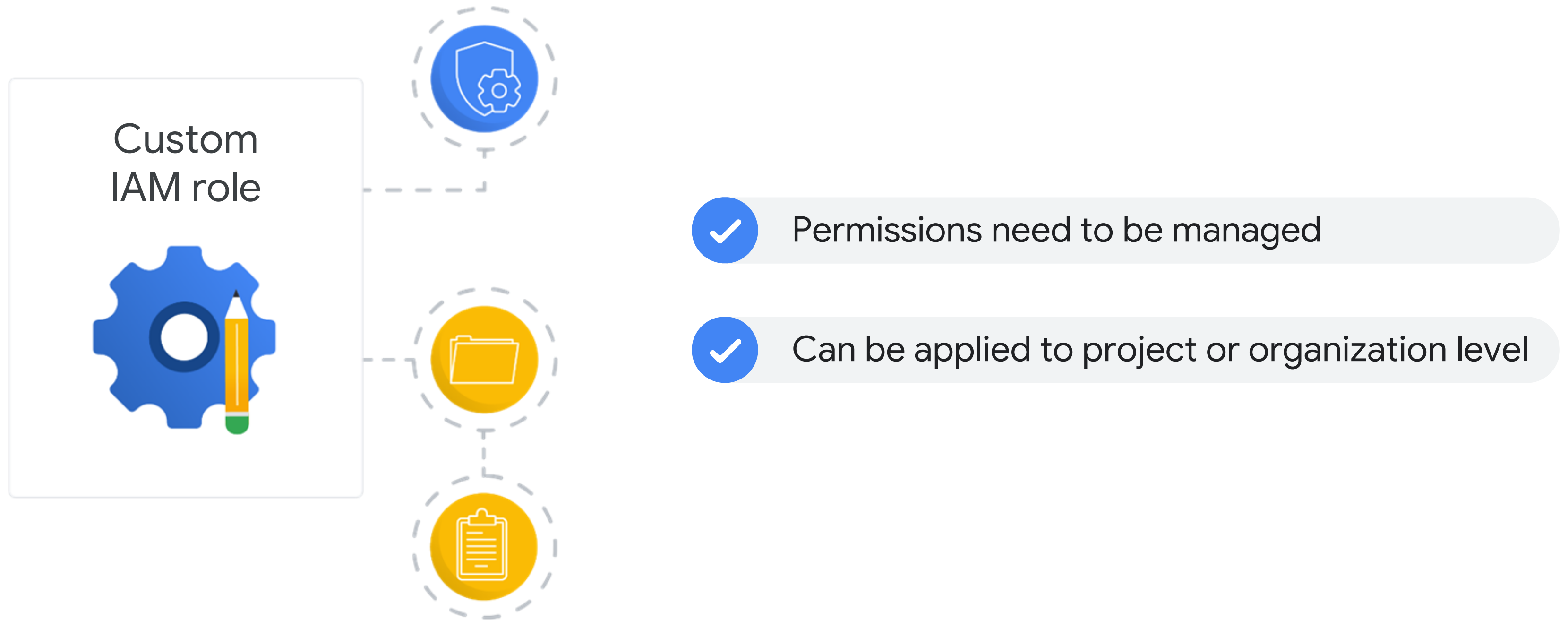
Predefined roles match job needs



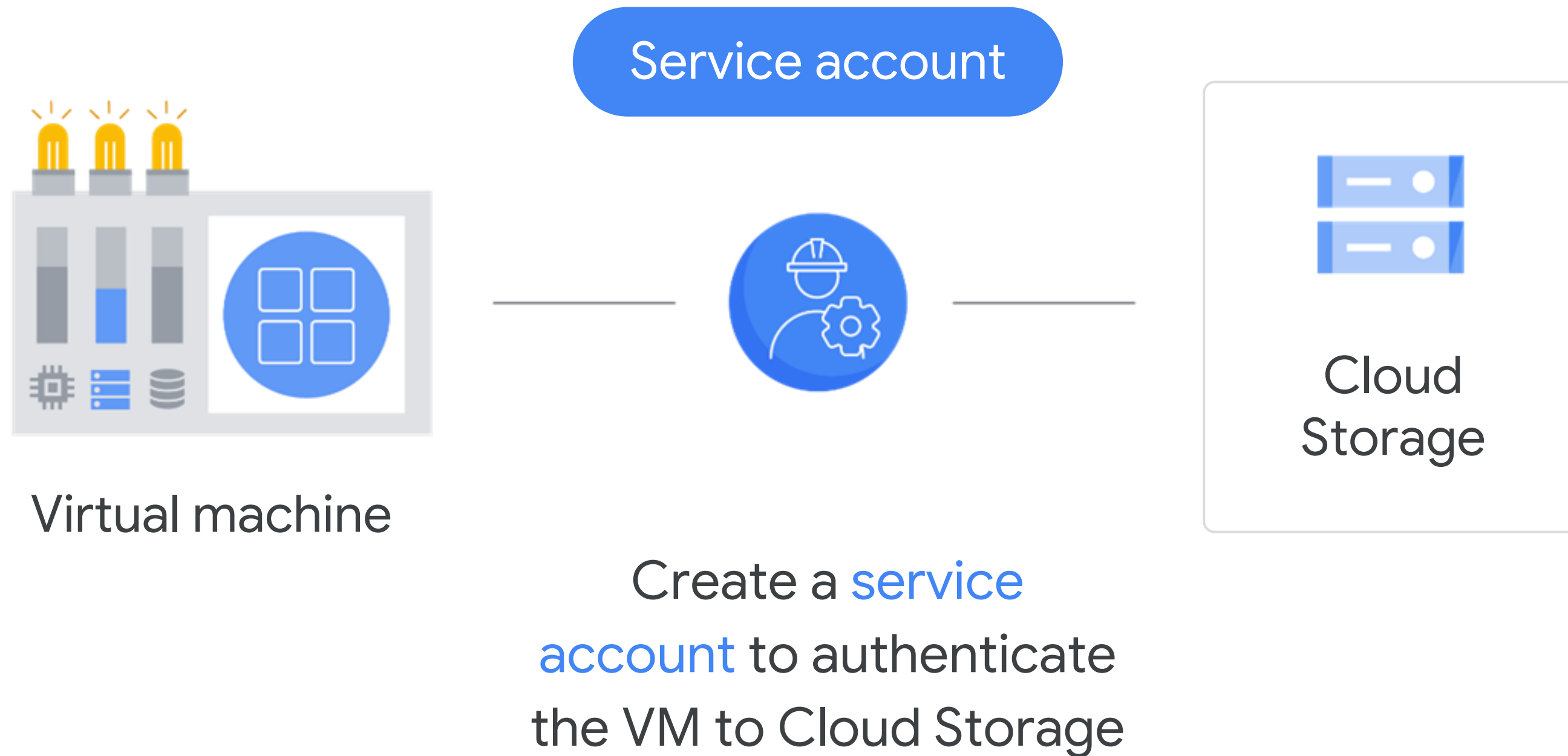
Custom roles are more specific and flexible



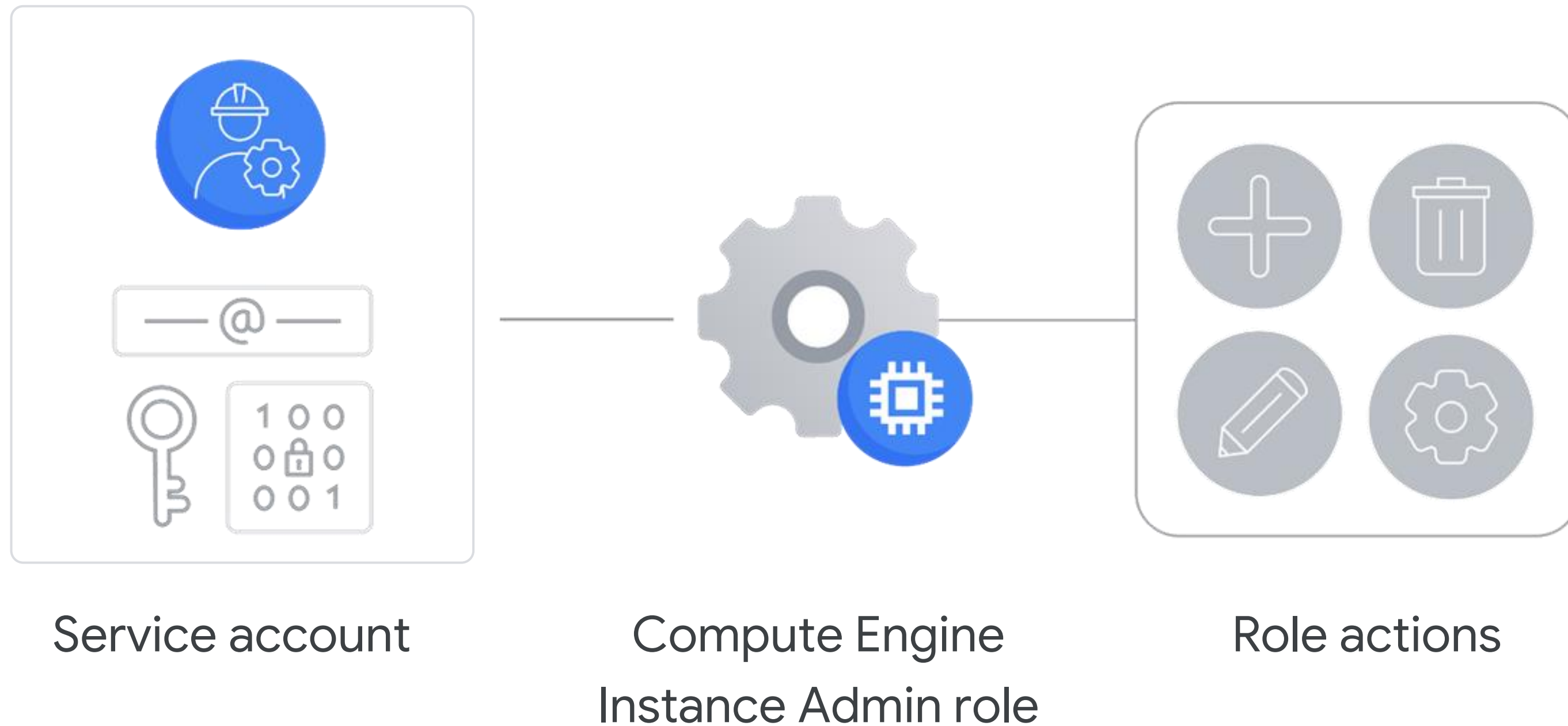
Custom roles are applied to projects and organizations



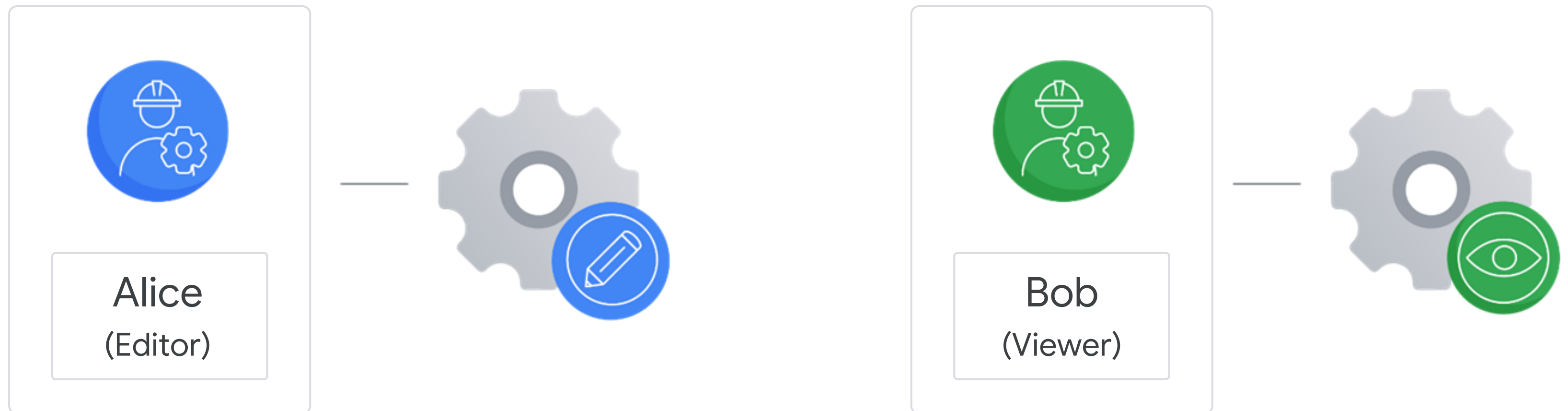
Permissions can be applied to service accounts



Service accounts are identified with email addresses



Service accounts are also managed by IAM



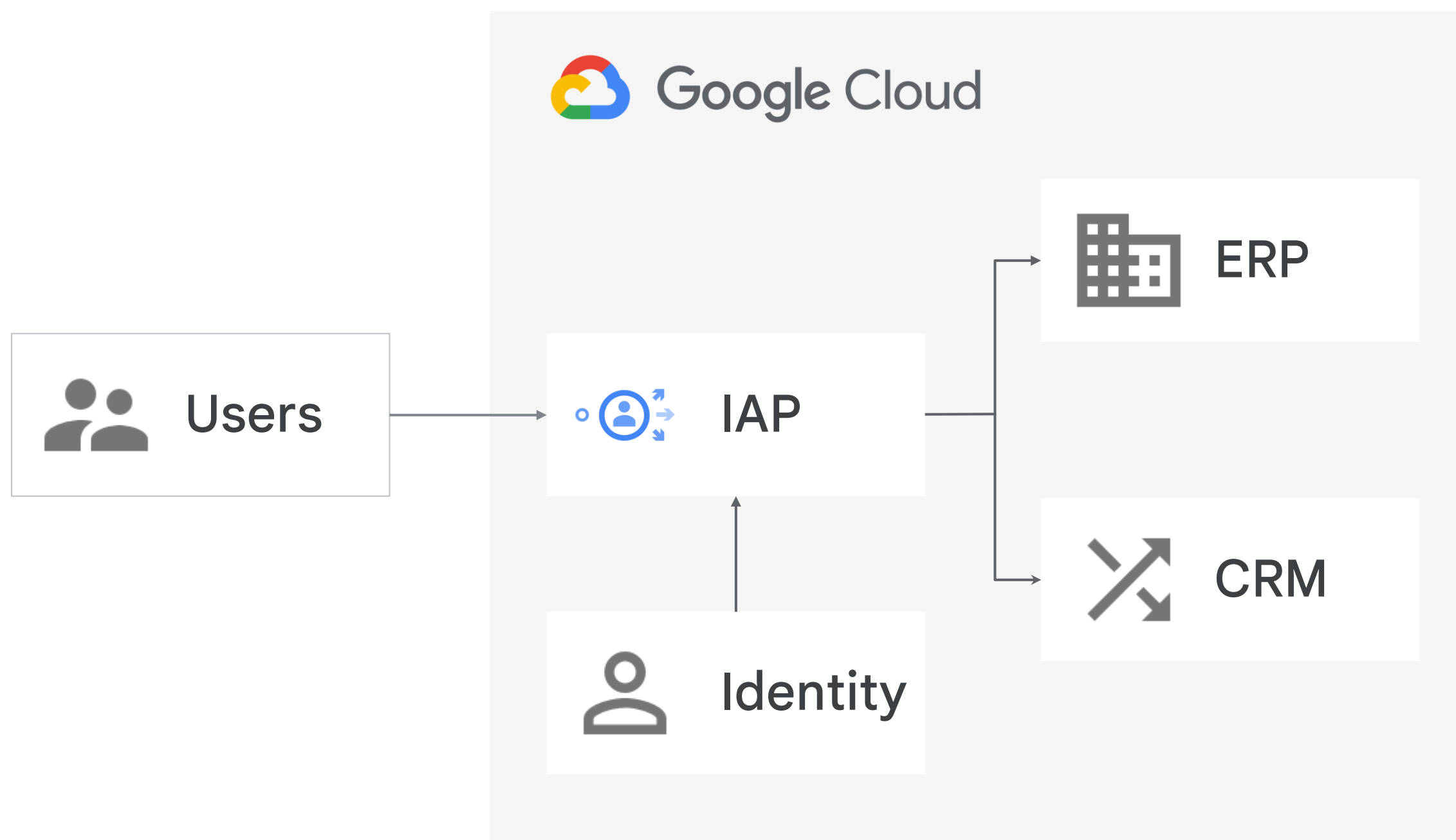


Agenda



- 01 Security in the cloud
- 02 The shared security model
- 03 Encryption options
- 04 Authentication and authorization with IAM
- 05 Lab: User Authentication: Identity-Aware Proxy**
- 06 IAM authorization best practices
- 07 Lab: IAM: Qwik Start
- 08 Quiz
- 09 Summary

Using Identity-Aware Proxy (IAP)

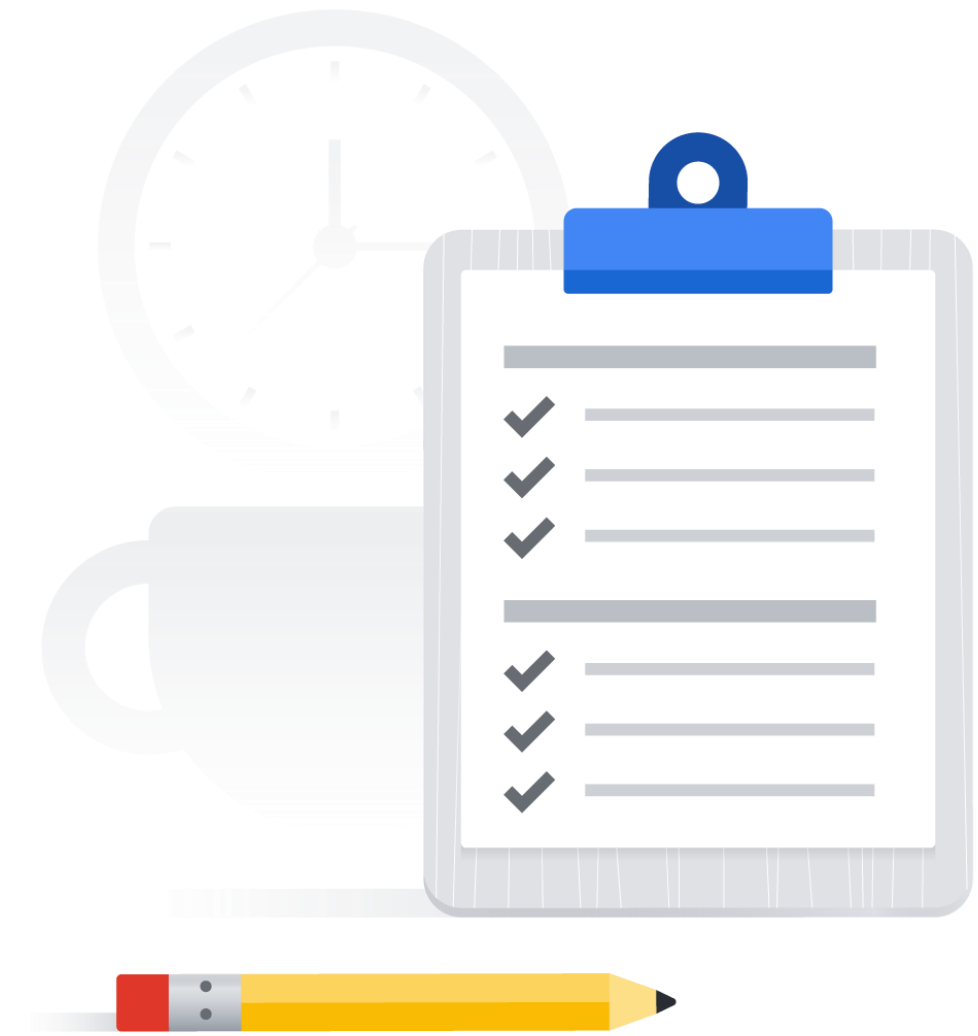


User Authentication: Identity-Aware Proxy

 Hands-on lab

During this lab, you'll:

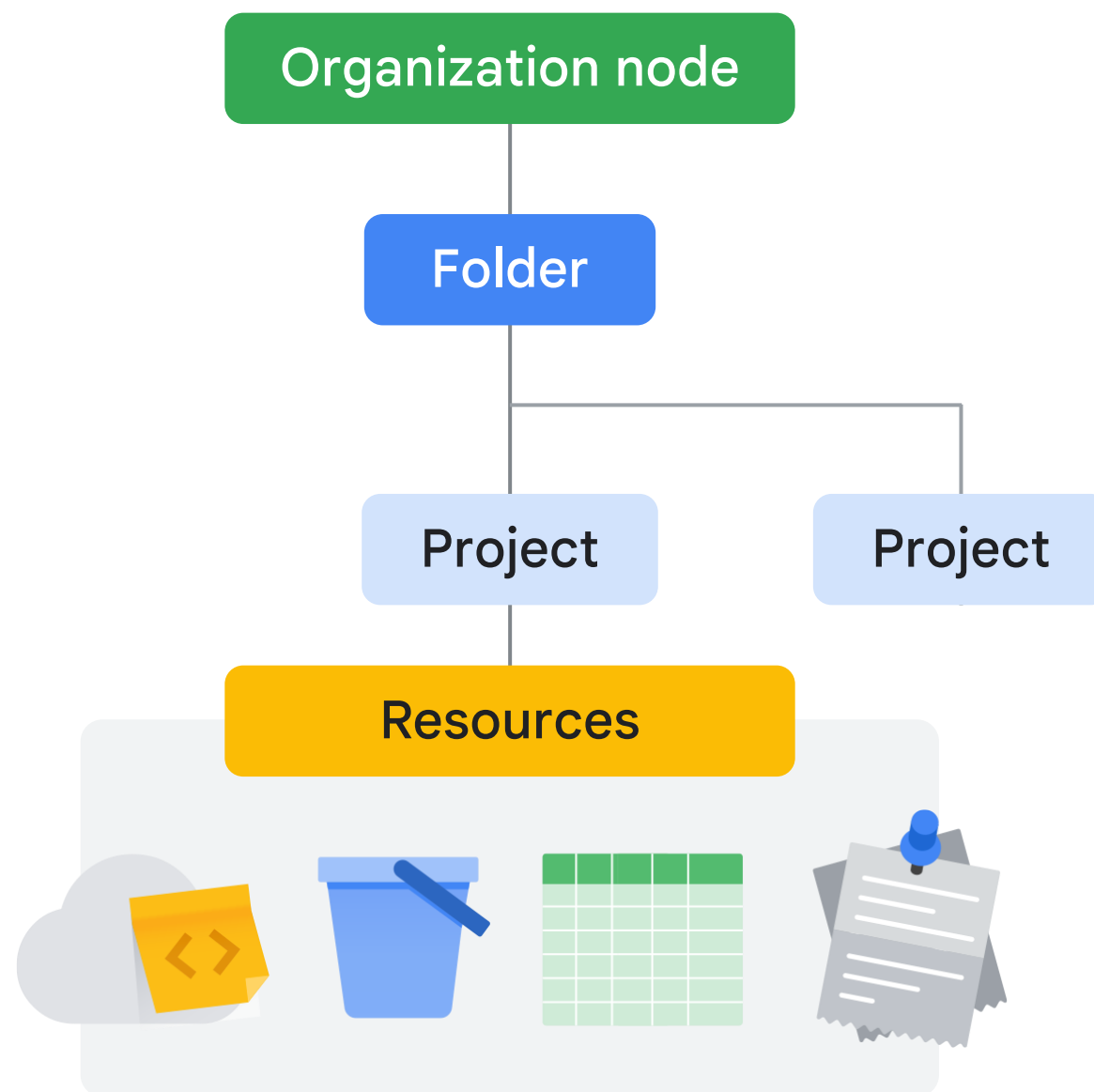
1. Use Python to write and deploy a simple App Engine app.
2. Enable and disable IAP to restrict access to your app.
3. Get user identify information from IAP into your app.
4. Cryptographically verify information from IAP to protect against spoofing.



Agenda

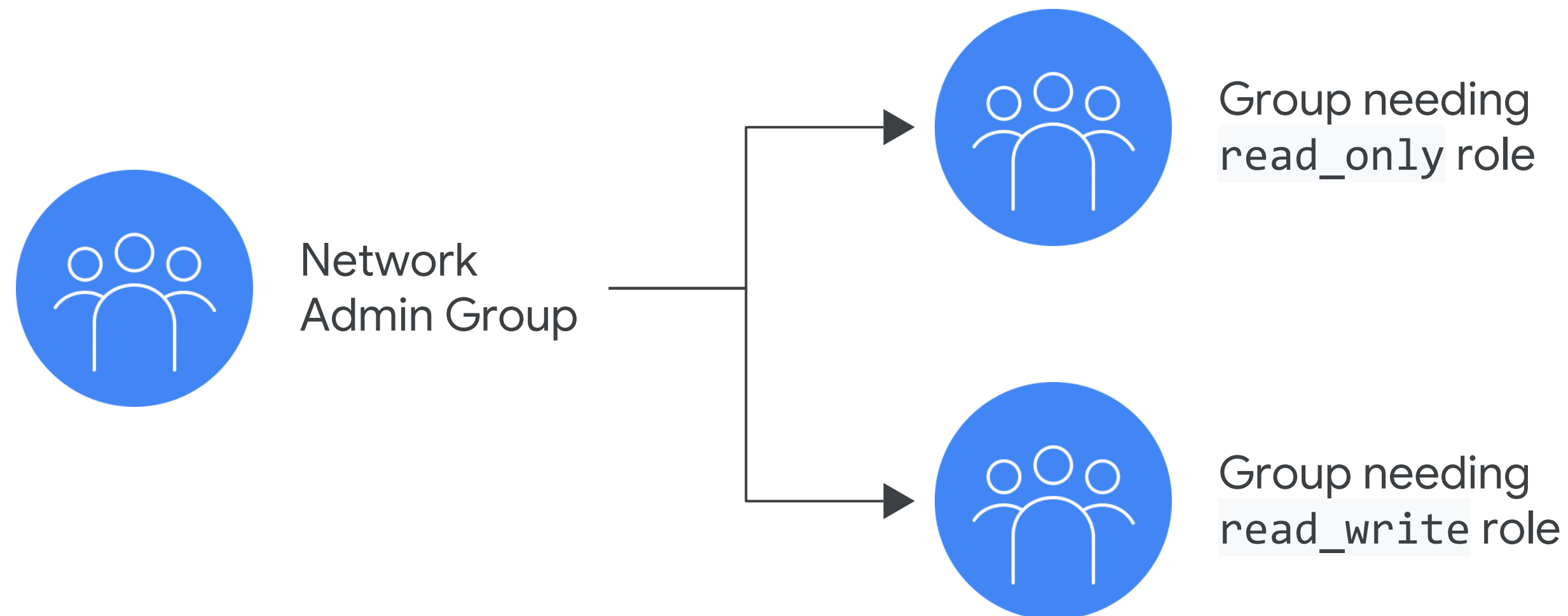
- 01 Security in the cloud
- 02 The shared security model
- 03 Encryption options
- 04 Authentication and authorization with IAM
- 05 Lab: User Authentication: Identity-Aware Proxy
- 06 IAM authorization best practices**
- 07 Lab: IAM: Qwik Start
- 08 Quiz
- 09 Summary

Leverage and understand the resource hierarchy

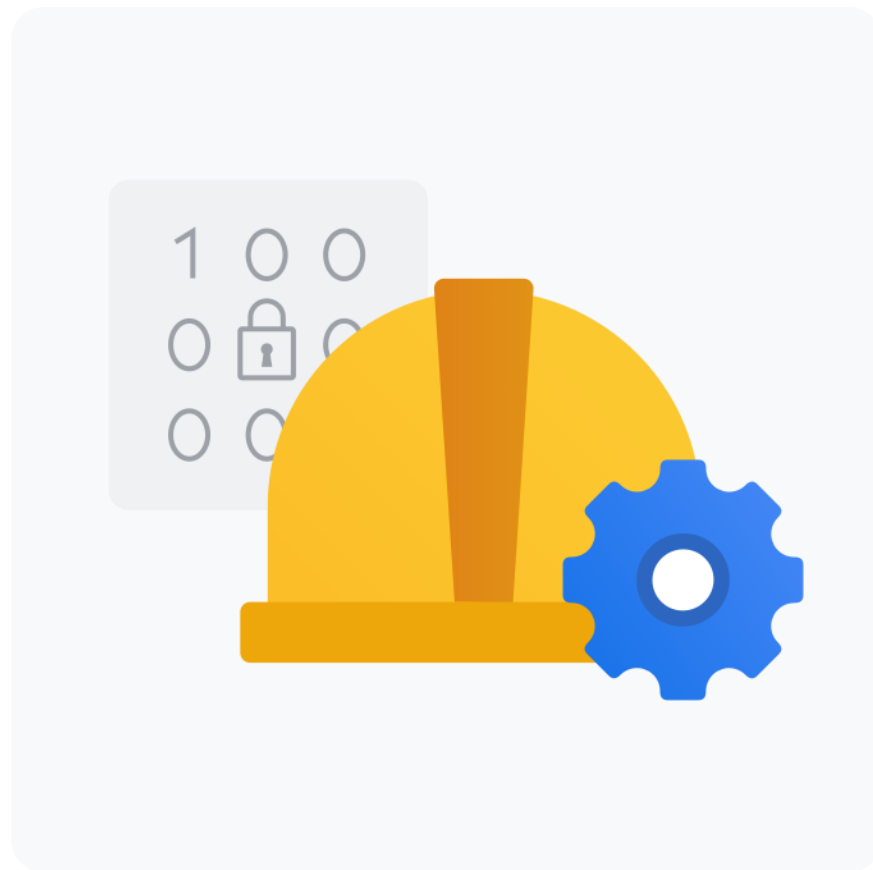


- ✓ Use projects to group resources
- ✓ Check the policy granted on each resource
- ✓ Use "principle of least privilege"
- ✓ Audit policies using Cloud Audit Logs
- ✓ Audit memberships of groups used in policies

Grant roles to groups instead of individuals



Best practices for service accounts



Use caution when granting the `serviceAccountUser` role.



Give a service account a display name that clearly identifies its purpose.



Establish a naming convention for service accounts.



Establish key rotation policies and methods.



Agenda



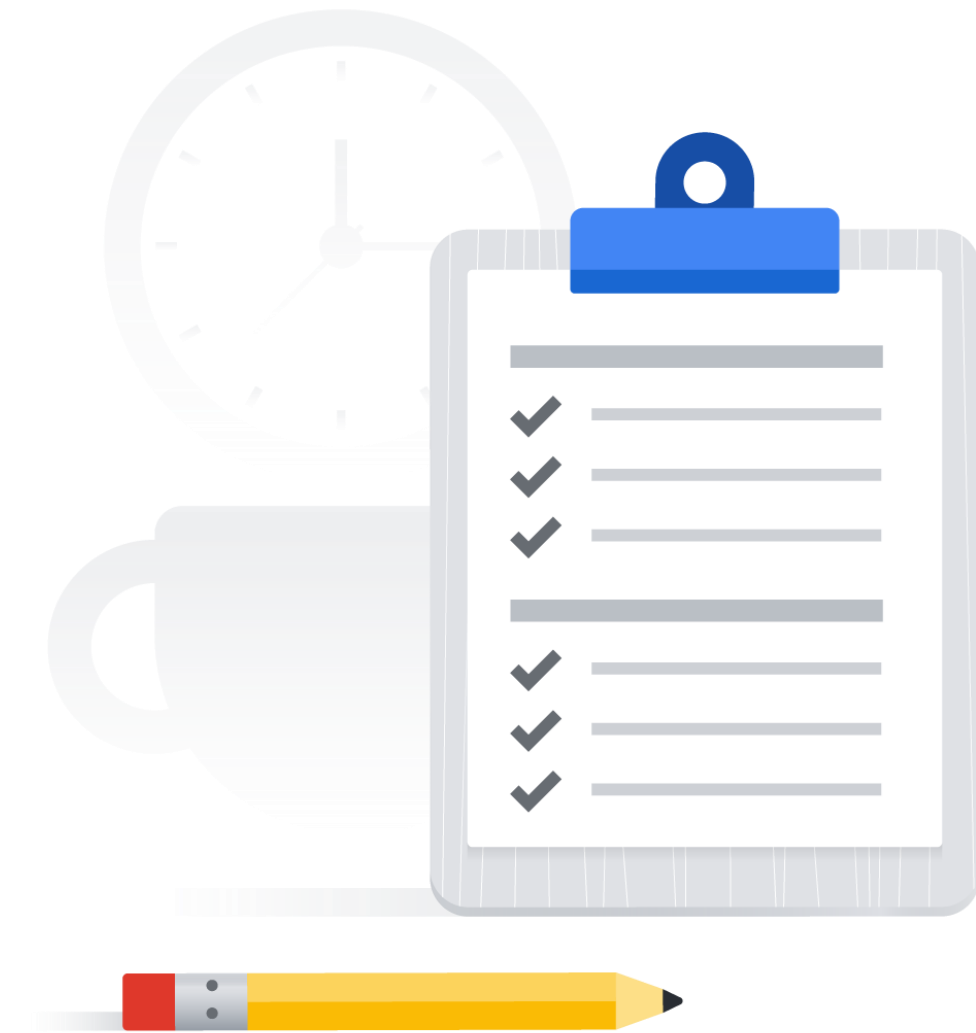
- | | |
|-----------|--|
| 01 | Security in the cloud |
| 02 | The shared security model |
| 03 | Encryption options |
| 04 | Authentication and authorization with IAM |
| 05 | Lab: User Authentication: Identity-Aware Proxy |
| 06 | IAM authorization best practices |
| 07 | Lab: IAM: Qwik Start |
| 08 | Quiz |
| 09 | Summary |

Lab: IAM: Qwik Start

 Hands-on lab

During this lab, you'll:

1. Explore editor roles.
2. Prepare a resource for access testing.
3. Remove project access.
4. Add storage permissions.
5. Verify access.





Agenda



- 01 Security in the cloud
- 02 The shared security model
- 03 Encryption options
- 04 Authentication and authorization with IAM
- 05 Lab: User Authentication: Identity-Aware Proxy
- 06 IAM authorization best practices
- 07 Lab: IAM: Qwik Start
- 08 Quiz**
- 09 Summary

Knowledge check

At which level of Google Cloud's infrastructure security will you find intrusion detection?

- A. The hardware infrastructure level
- B. The storage services level
- C. The internet communication level
- D. The operational security level



Knowledge check

At which level of Google Cloud's infrastructure security will you find intrusion detection?

- A. The hardware infrastructure level
- B. The storage services level
- C. The internet communication level
- D. The operational security level



Knowledge check

When a customer moves an application to Google Cloud, which one of the following does the customer remain responsible for?

- A. Network integrity
- B. Disk encryption
- C. Data security
- D. Physical security



Knowledge check

When a customer moves an application to Google Cloud, which one of the following does the customer remain responsible for?

A. Network integrity

B. Disk encryption

C. Data security

D. Physical security



Knowledge check

With which encryption option does a customer encrypt data before sending it to Google Cloud?

- A. Client-side encryption
- B. Customer-managed encryption keys (CMEK)
- C. Google encryption by default
- D. Customer-supplied encryption Keys (CSEK)



Knowledge check

With which encryption option does a customer encrypt data before sending it to Google Cloud?

- A. Client-side encryption
- B. Customer-managed encryption keys (CMEK)
- C. Google encryption by default
- D. Customer-supplied encryption Keys (CSEK)



Knowledge check

Which IAM role is the most broad in scope?

- A. Basic
- B. Predefined
- C. Custom



Knowledge check

Which IAM role is the most broad in scope?

A. Basic

B. Predefined

C. Custom





Agenda



- 01 Security in the cloud
- 02 The shared security model
- 03 Encryption options
- 04 Authentication and authorization with IAM
- 05 Lab: User Authentication: Identity-Aware Proxy
- 06 IAM authorization best practices
- 07 Lab: IAM: Qwik Start
- 08 Quiz
- 09 Summary**

Summary

- ✓ Outlined how security is administered in Google Cloud.
- ✓ Explored the different encryption options with Google Cloud.
- ✓ Identified best practices when configuring authentication and authorization by using Identity and Access Management (IAM).



